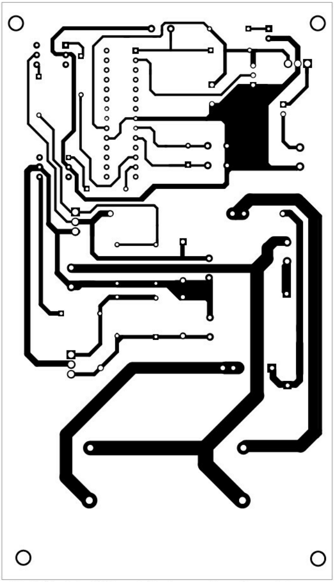
Fig. 7: Actual-size PCB layout of the fan speed increasing regulator

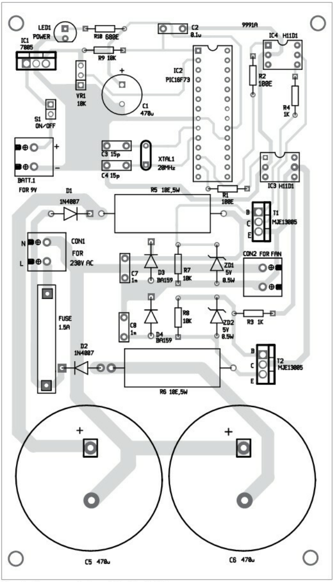

Fig. 8: Components layout for the PCB

PWM pulses are fed to H11D1 optoisolators. For ceiling and table fan loads, if H11D1 optoisolator is used, switching transistors can be MJE13005 or 3039.

The upper section of fan speed controller circuit (shown in Fig. 6) is safe as it works on low voltage only. But after mains supply is given to the bridge, it is dangerous to touch any part of the circuit. Turn on the circuit using VR1 with its knob.

In Fig. 6, the bottom section of the circuit shows the connection to 230V AC mains power. 1N4007 diodes are used for the fan load as these have high inductance. The circuit is very simple to wire. You can even assemble it on a breadboard as transistors are 3-pin TO220 type. Use suitable heat-sinks for transistors T1 and T2.

After wiring the circuit, switch on the AC mains power. It is advisable to use a variable auto transformer first and vary the voltage slowly from 140V. Otherwise, the fuse may blow up at startup if there is any wiring fault.

Connect the fan to the circuit through a suitable connector. For a ceiling fan already fitted in your room, the connections from the fan should be isolated and brought to the circuit through a long twin-flex wire for testing.

The fan must run at the speed controlled by VR1. Fan speed increases in eight steps through VR1 adjustment. VR1 should be set to the maximum value first to get a slower speed for start. Then it can be brought downwards to increase the fan speed.

It's not possible to observe the output voltage waveform on a scope, as the oscilloscope's earth point should not receive hot volt-